Asteroseismology with GYRE

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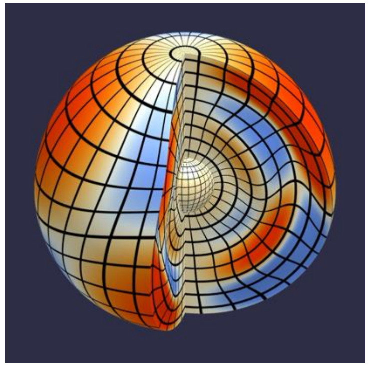
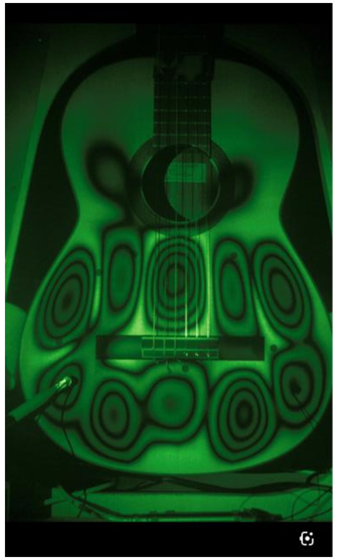
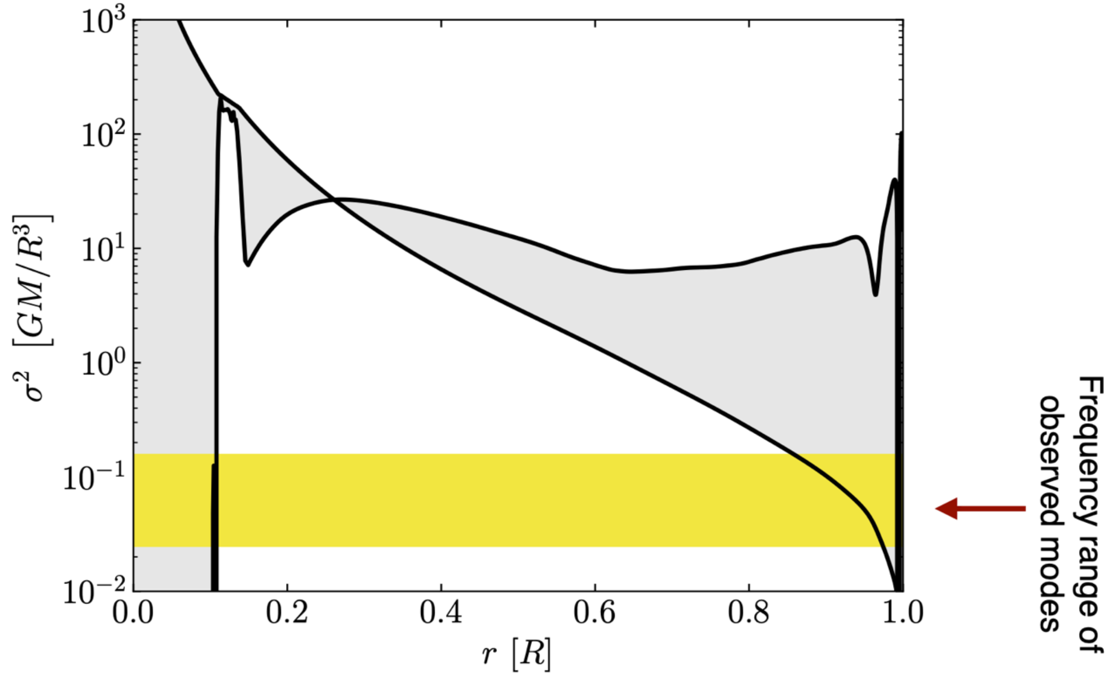
Asteroseismology: Probing Stellar Interiors

- observed frequencies → stellar interior → stellar parameters







- density, temperature, pressure profile
- rotation
- mixing

……

Physics of Stellar Oscillations

- Perturbation theory, fluid dynamics and thermal physics.
- Then the perturbed fluid equations (for non-rotating stars) are:

- Continuity

(after integrating with respect to time)

$$\rho' + \mathrm{div}\left(\rho_0 \boldsymbol{\delta r}\right) = 0$$

- Equation of motion

$$\rho_0 \frac{\partial^2 \boldsymbol{\delta r}}{\partial t^2} = \rho_0 \frac{\partial \boldsymbol{v}}{\partial t} = -\nabla p' + \rho_0 \boldsymbol{g}' + \rho' \boldsymbol{g}_0$$

pressure gravity buoyancy

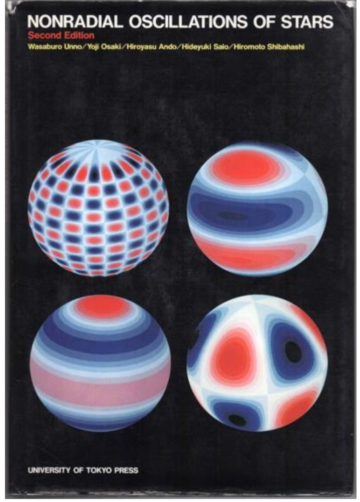
- Poisson

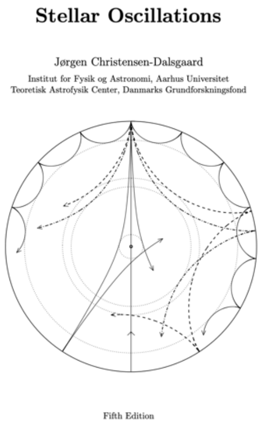
$$\nabla^2 \Phi' = 4\pi G \rho'$$

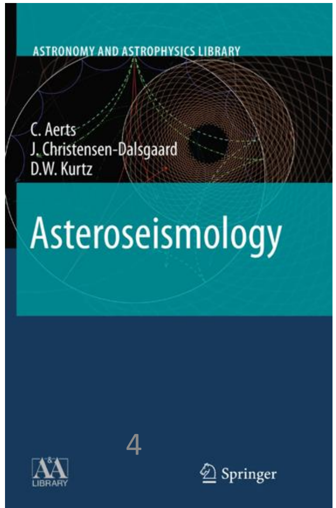
potential field density source distribution

- Energy equation

$$p' + \boldsymbol{\delta r} \cdot \nabla p_0 = \frac{\Gamma_{1,0} p_0}{\rho_0} \left(\rho' + \boldsymbol{\delta r} \cdot \nabla \rho_0\right)$$







GYRE solves for Free Oscillations

- Free oscillations: $\underbrace{\mathcal{L}(\xi)}_{\text{linear operator}} + \sigma^2 \xi = 0$

GYRE

- Solving this eigenvalue problem gives: eigenfrequencies $\sigma_{n\ell}$, eigenmodes $\xi_{n\ell}(r)$

Finding the Beat

Using GYRE on a MESA model to calculate high-order g-mode period spacings

Build GYRE compile + test → **Load MESA model** FGONG profile → **Choose g modes** ℓ=1, m=0, n<0 → **Scan frequency** inverse grid → **Plot ΔΠ** nearly constant

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